How Income Relates to Restaurant Spending

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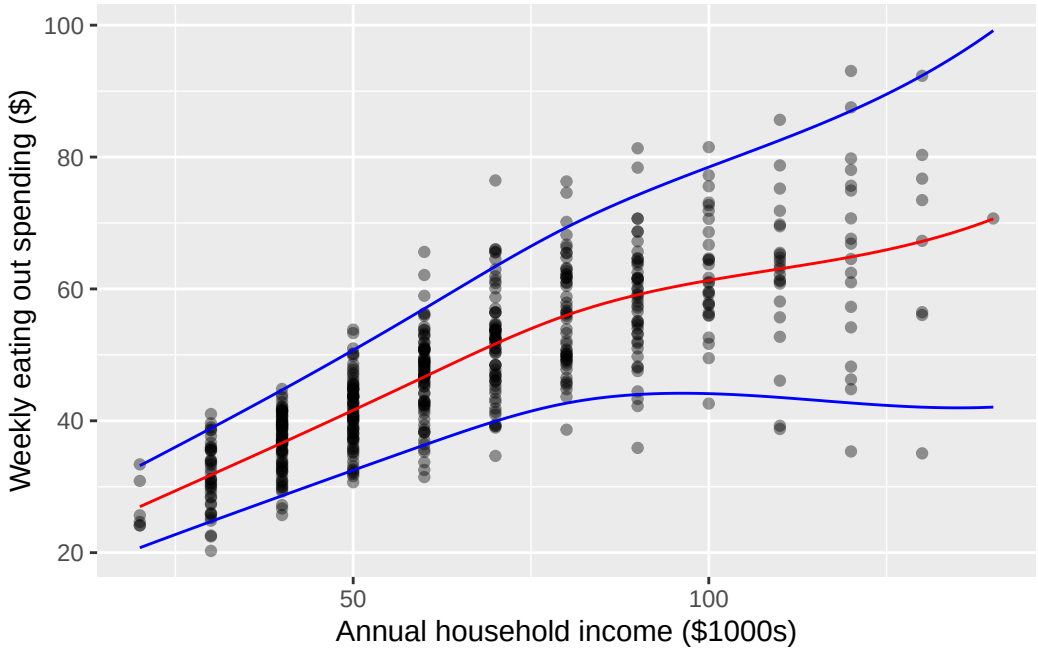
Introduction

Eating out is something that people often spend more on when their income rises. The National Restaurant Association wants to know how much more which is why they reached out to us. The answer is important for projecting food demand, for judging whether owning a restaurant is still a good career move as incomes grow, and for understanding how much of a household's food budget changes from cooking at home to paying for quality and convenience as income increases.

The data come from the Food Demand Survey (FooDS) and includes 523 households. For each household we have annual income (in thousands of dollars) and average weekly spending on food away from home (in dollars).

Main Conclusions

Eating out spending does go up with income, but not at a steady rate. It climbs quickly through the lower and middle income range and then levels off at higher incomes. High-income households are also much less predictable. Two households earning $100,000 can differ in weekly eating out spending far more than two households earning $40,000.



The red line is the model's fitted average weekly spending, and the blue lines are a 90% prediction band for a single household. The band gets wider as income grows this is the increase in variability referenced before. We are able to estimate more accurately for households with lower income than those with higher income.

The advisers guessed that a "healthy" restaurant economy should add about $0.50 or more of weekly spending for each $1,000 of income. To get one overall number to compare against the benchmark, we fit the same model with a straight-line mean: the income effect comes out to about $0.44 per $1,000, with a 95% confidence interval of (0.42, 0.47). Since the whole interval is below $0.50, we believe that this is not a healthy economy. The curve however is not linear and suggests that the effect is strongest in the lower and middle income range and weaker at high incomes, so the benchmark comes closest to being met for lower income households.

If incomes keep growing 3-4% per year, a household's income in five years will be roughly 1.16 to 1.22 times what it is today. The table below shows what the model predicts for weekly eating out spending now versus five years from now for each of the growth rates (3% per year and 4% per year), including how the spread (standard deviation) of spending grows along with the average.

	Income	mean_now	sd_now	mean_5yr_3pct	sd_5yr_3pct	mean_5yr_4pct	sd_5yr_4pct
1	40	36.65226	4.876583	39.80897	5.289339	41.00538	5.452976
2	60	46.66520	6.286136	51.48192	7.100884	53.16947	7.432941
3	80	56.00441	8.103113	59.79959	9.532866	60.83268	10.131828
4	100	61.29794	10.445278	64.08979	12.797776	65.26356	13.810676

For example, a household now earning $60,000 is predicted to spend about $46.7 per week eating out, give or take about $6.30. In five years at the same point in the income distribution, that rises to about $53.20 per week, give or take about $7.40. So rising incomes should bring both more restaurant spending and a wider range of spending across households. The largest projected incomes go past what was observed in the survey, so the high-income projections are not as reliable since they need the current pattern to hold but we don't know if it will. These predictions are still good for the lower incomes and imply a growth across all income levels.

Modeling Details

Why standard regression is not enough

A standard regression of EatingOut on Income gives one slope, about $0.39 per $1,000, and assumes both a straight line relationship and the same spread at every income. Neither of these assumptions are true with our data. Running a test for equal variance (a Breusch-Pagan test, p-value < 0.001) shows us that the variance of spending clearly changes with income. Another test that checks whether the errors look normal (a Kolmogorov-Smirnov test, p-value $= 0.04$) tells us that using a straight line may not be a great option. Standard regression can't describe a relationship whose spread changes with income, so a different kind of model is needed.

The advanced model

We used a model from the GAMLSS family that lets both the center and the spread of spending depend on income. Each household's weekly spending is modeled as

$$Y_i \sim N(\mu_i, \sigma_i^2),$$

where the average spending μ_i is a smooth, flexible curve of income instead of a straight line. This curved line is called a spline and it is a few simple cubic curves joined at break points on the income axis (we used $45,000, $60,000, and $80,000). This lets the income effect be steep where spending responds strongly and flat where it levels off, all as part of one regression model.

The spread is modeled at the same time. The variance follows

$$\log \sigma_i^2 = 2(\theta_0 + \theta_1 \,\text{Income}_i),$$

so the standard deviation can grow or shrink smoothly with income, and the log keeps the variance positive. Everything is estimated by maximum likelihood: the computer repeatedly adjusts the coefficients until the model is as likely as possible given the data.

The fitted variance coefficient is $\hat{\theta}_1 \approx 0.0127$, with a 95% confidence interval of (0.0104, 0.0150). Since the whole interval is above zero, spread really does grow with income and each extra $1,000 of income multiplies the standard deviation of weekly spending by about 1.013, roughly a 1.3% increase. A simpler regression model cannot represent these nuances in the data.

This model is also what produces the prediction line in the figure above. Because each income has its own mean and standard deviation, a 90% interval for one household is just $\mu_i \pm 1.645\,\sigma_i$, and it widens at higher incomes.

Model validation

If the model is right, we expect the residuals standardized by each household's predicted spread to look like standard normal noise and they do. The Kolmogorov-Smirnov test gives a p-value of 0.545, so the standardized residuals match a normal distribution, and a Breusch-Pagan test on them gives a p-value of 1, so our model is properly accounting for the increase in variance. The model explains about 64.8% of the variation in weekly eating out spending.

We also cross-validated the predictions the data was split into 25 groups (called folds), then for each fold, we keep it out and train the model on the rest of the data, then we check to see how well that model can predict the data we kept out. We then look at the average error to see how well our model works. The average prediction error (RMSE) was about $7.53 per week, about the same as the standard regression ($7.70). The main difference is that we can trust this model's uncertainty. Its 90% prediction intervals captured 88.3% of households that it wasn't trained on, very close to the advertised 90%.

Appendix: Code

```
# Load packages and data
library(tidyverse)
library(splines)
library(gamlss)

food <- vroom::vroom("../DATA/FoodExpenses.txt")
food_gamlss <- gamlss(
    EatingOut ~ bs(Income, knots = c(45, 60, 80), degree = 3),
    sigma.formula = ~Income,
    family = NO,
    data = food,
    trace = FALSE
)
```

```r
beta_hat <- coef(food_gamlss, what = "mu")
theta_hat <- coef(food_gamlss, what = "sigma")

# 95% CI for the variance coefficients
theta_ci <- confint(food_gamlss, what = "sigma")

# Prediction band
income_grid <- data.frame(
    Income = seq(min(food$Income), max(food$Income), length.out = 500)
)
income_grid <- income_grid %>%
    mutate(
        fit = predict(food_gamlss, what = "mu", newdata = ., data = food),
        sigma = predict(food_gamlss, what = "sigma", type = "response",
                        newdata = ., data = food),
        lwr = fit - qnorm(0.95) * sigma,
        upr = fit + qnorm(0.95) * sigma
    )

fit_plot <- ggplot(food, aes(x = Income, y = EatingOut)) +
    geom_point(alpha = 0.4) +
    geom_line(data = income_grid, aes(y = fit), color = "red") +
    geom_line(data = income_grid, aes(y = lwr), color = "blue") +
    geom_line(data = income_grid, aes(y = upr), color = "blue") +
    labs(
        x = "Annual household income ($1000s)",
        y = "Weekly eating out spending ($)"
    )

fit_plot
# Income effect
income_gamlss <- gamlss(
    EatingOut ~ Income,
    sigma.formula = ~Income,
    family = NO,
    data = food,
    trace = FALSE
)
income_effect <- coef(income_gamlss, what = "mu")["Income"]
income_effect_ci <- confint(income_gamlss, what = "mu")["mu.Income", ]
# 5-year projection
proj_incomes <- data.frame(Income = c(40, 60, 80, 100))
```

```r
incomes_3pct <- data.frame(Income = proj_incomes$Income * 1.16)
incomes_4pct <- data.frame(Income = proj_incomes$Income * 1.22)

mu_now <- predict(food_gamlss, what = "mu",
                  newdata = proj_incomes, data = food)
sd_now <- predict(food_gamlss, what = "sigma", type = "response",
                  newdata = proj_incomes, data = food)

mu_5yr_3pct <- predict(food_gamlss, what = "mu",
                       newdata = incomes_3pct, data = food)
sd_5yr_3pct <- predict(food_gamlss, what = "sigma", type = "response",
                       newdata = incomes_3pct, data = food)

mu_5yr_4pct <- predict(food_gamlss, what = "mu",
                       newdata = incomes_4pct, data = food)
sd_5yr_4pct <- predict(food_gamlss, what = "sigma", type = "response",
                       newdata = incomes_4pct, data = food)

projection_table <- data.frame(
    Income = proj_incomes$Income,
    mean_now = mu_now, sd_now = sd_now,
    mean_5yr_3pct = mu_5yr_3pct, sd_5yr_3pct = sd_5yr_3pct,
    mean_5yr_4pct = mu_5yr_4pct, sd_5yr_4pct = sd_5yr_4pct
)
projection_table
# Standard regression
food_lm <- lm(EatingOut ~ Income, data = food)
lm_slope <- coef(food_lm)["Income"]

# Assumption checks
standardized_resids_lm <- MASS::stdres(food_lm)
lm_bp <- lmtest::bptest(food_lm)
lm_ks <- ks.test(standardized_resids_lm, "pnorm", mean = 0, sd = 1)
# Model validation
zres <- residuals(food_gamlss, what = "z-scores")
zres_ks <- ks.test(zres, "pnorm", mean = 0, sd = 1)
resid_dataframe <- food %>% mutate(r = zres)
zres_bp <- lmtest::bptest(formula = r ~ Income, data = resid_dataframe)

r_squared <- 1 -
    sum((food$EatingOut - fitted(food_gamlss, what = "mu"))^2) /
    sum((food$EatingOut - mean(food$EatingOut))^2)
```

```r
# Cross-validation
set.seed(42)
K <- 25
folds <- sample(rep(1:K, length.out = nrow(food)))

cross_validate <- function(fold_num) {
    train <- food[folds != fold_num, ]
    test <- food[folds == fold_num, ]

    train_gamlss <- gamlss(
        EatingOut ~ bs(Income, knots = c(45, 60, 80), degree = 3),
        sigma.formula = ~Income,
        family = NO,
        data = train,
        trace = FALSE
    )
    mu_pred <- predict(train_gamlss, what = "mu", newdata = test, data = train)
    sigma_pred <- predict(
        train_gamlss, what = "sigma", type = "response",
        newdata = test, data = train
    )

    train_lm <- lm(EatingOut ~ Income, data = train)
    lm_pred <- predict(train_lm, newdata = test)

    c(
        gamlss_rmse = sqrt(mean((test$EatingOut - mu_pred)^2)),
        coverage = mean(
            test$EatingOut >= mu_pred - qnorm(0.95) * sigma_pred &
                test$EatingOut <= mu_pred + qnorm(0.95) * sigma_pred
        ),
        lm_rmse = sqrt(mean((test$EatingOut - lm_pred)^2))
    )
}

cv_results <- sapply(1:K, cross_validate)
cv_gamlss_rmse <- mean(cv_results["gamlss_rmse", ])
cv_coverage <- mean(cv_results["coverage", ])
cv_lm_rmse <- mean(cv_results["lm_rmse", ])
```